CLASS 9
1.2.5- EVALUATION

Q1. What is Evaluation?

Evaluation is the process of checking how reliable an AI model is. We feed test data into the model and compare its predictions with the actual answers. It's important to use fresh test data, not the training data, otherwise the model may just memorize answers (overfitting).

Q2. What are various Model evaluation techniques?

- Accuracy – how many predictions were correct overall
- Precision – how many predicted positives were actually correct
- Recall – how many actual positives were correctly predicted
- F1 Score – a balance between Precision and Recall

Q3. Why is model evaluation important in AI projects?

Evaluation helps us test different models and choose the best one. It shows whether the model is solving the identified problem correctly, checks its performance on new data, and helps identify areas for improvement before the model is actually used in real life.

Q4. What do you understand by the terms True Positive and False Positive?

Basis	True Positive	False Positive
Meaning	Prediction matches reality – both say Yes	Prediction says Yes but reality is No
Accuracy	Correct prediction	Incorrect prediction
Example (Forest Fire)	Model predicts fire, and there is actually a fire	Model predicts fire, but there is no fire

CLASS 9
1.2.6- DEPLOYMENT

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1. Does modeling mean creating an AI model?

Answer: (b) No – Modelling means developing algorithms using existing/new data; it is a stage in the cycle, not just "creating."

2. Can we use AI on mobile phones?

Answer: (a) Yes

3. What is deployment in the context of an AI project cycle?

Answer: Deployment is the final stage of the AI project cycle where the tested AI model or solution is implemented and made available for real-world use, such as through a mobile app or website.

4. Why is deployment an important phase in the AI project cycle?

Answer: Deployment makes the AI solution usable by real people to solve the actual problem. Without deployment, even a well-trained model stays unused and cannot benefit users in real life.

5. What are some common challenges in deploying AI models?

Answer: Common challenges include integrating the model with existing systems, ensuring it works accurately on real-world data, ongoing monitoring and maintenance, technical/infrastructure limitations, and making sure users can easily access and use it.

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1. Rearrange the steps of AI project cycle in correct order:

Answer: b → a → d → c → f → e (Problem Scoping → Data Acquisition → Data Exploration → Modelling → Evaluation → Deployment)

2. The process of breaking down the big problem into a series of simple steps is known as:

Answer: (b) Modularity

3. The primary purpose of data exploration in AI project cycle is:

Answer: (c) To discover patterns and insights in data

4. Deployment is the final stage in the AI project cycle where the AI model or solution is implemented in a real-world scenario. (True/False)

Answer: True

5. Identify A, B and C in the following diagram (Hint: How AI, ML & DL related to each other):

Answer: A = Artificial Intelligence, B = Machine Learning, C = Deep Learning